

Package ‘RCamelsPE’

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Type Package

Title Tools for Reading, Processing and Visualizing CAMELS-PE Data

Version 0.1.0

Description Provides tools to efficiently read, process and visualize hydrological, climatic, physiographic and geospatial data from the CAMELS-PE dataset (Catchment Attributes and Meteorology for Large-sample Studies - Peru). The package does not include the dataset and is designed to work with externally stored CAMELS-PE data. Functions support reading time series, catchment attributes, metadata, and geospatial data, as well as basic visualization of time series and spatial patterns.

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Encoding UTF-8

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Depends R (>= 4.1)

Imports dplyr,
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ggplot2,
sf,
terra,
rlang

Suggests testthat (>= 3.0.0),
knitr,
rmarkdown,
pkgdown

Config/testthat/edition 3

URL <https://github.com/yourusername/RCamelsPE>

BugReports <https://github.com/yourusername/RCamelsPE/issues>

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get_camels_path	<i>Get CAMELS-PE dataset path</i>
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Description

Get CAMELS-PE dataset path

Usage

get_camels_path()

Value

Character path to the CAMELS-PE root directory.

plot_attribute_map	<i>Plot CAMELS-PE Attribute Map</i>
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Description

Creates a thematic map of CAMELS-PE catchments using a selected attribute. The function joins attribute data with catchment geometries by gauge_id and returns a ggplot object.

Usage

```
plot_attribute_map(  
  catchments,  
  attributes,  
  variable,  
  gauges = NULL,  
  na_color = "grey80",  
  ...  
)
```

Arguments

catchments	An sf object with catchment polygons.
attributes	A data frame or tibble with CAMELS-PE attributes.
variable	Character string. Name of the attribute to visualize.
gauges	Optional sf object with gauge point locations.
na_color	Character string. Color for missing values.
...	Additional arguments passed to <code>ggplot2::geom_sf()</code> for catchments.

Value

A ggplot object.

Examples

```
## Not run:
catchments <- read_geospatial("catchments")
gauges <- read_geospatial("gauges")
attr <- read_attributes("topographic")

plot_attribute_map(
  catchments = catchments,
  attributes = attr,
  variable = "area_km2",
  gauges = gauges
)

## End(Not run)
```

plot_catchments	<i>Plot CAMELS-PE Catchments</i>
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Description

Creates a simple map of CAMELS-PE catchments and, optionally, gauge locations. This function is useful for quick inspection of the spatial distribution of catchments and stations.

Usage

```
plot_catchments(catchments, gauges = NULL, fill = NULL, ...)
```

Arguments

catchments	An sf object with catchment polygons, typically obtained with <code>read_geospatial("catchments")</code> .
gauges	Optional sf object with gauge point locations, typically obtained with <code>read_geospatial("gauges")</code> .
fill	Optional character string. Name of a catchment column used to fill polygons. If NULL, catchments are plotted with a constant fill.
...	Additional arguments passed to <code>ggplot2::geom_sf()</code> for catchments.

Value

A ggplot object.

Examples

```
## Not run:
catchments <- read_geospatial("catchments")
gauges <- read_geospatial("gauges")

plot_catchments(catchments)
plot_catchments(catchments, gauges)

## End(Not run)
```

plot_timeseries	<i>Plot CAMELS-PE Time Series</i>
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Description

Creates a time series plot for one CAMELS-PE variable, such as precipitation, observed streamflow, or temperature. The function returns a ggplot object, so it can be further customized using standard ggplot2 layers.

Usage

```
plot_timeseries(
  data,
  variable = "flow_obs",
  gauge_id = NULL,
  date_col = "date",
  facet = TRUE,
  scales = "free_y",
  ...
)
```

Arguments

data	A data frame or tibble containing CAMELS-PE time series.
variable	Character string. Name of the variable to plot.
gauge_id	Optional character vector. Gauge IDs to filter before plotting.
date_col	Character string. Name of the date column.
facet	Logical. If TRUE, creates one panel per gauge ID.
scales	Character string. Scales passed to facet_wrap().
...	Additional arguments passed to ggplot2::geom_line().

Value

A ggplot object.

Examples

```
## Not run:
ts <- read_timeseries(
  gauge_id = c("PE_0001", "PE_0002"),
  vars = c("date", "gauge_id", "prec", "flow_obs", "tmean")
)

plot_timeseries(ts, variable = "flow_obs")
plot_timeseries(ts, variable = "prec", linewidth = 0.4)
plot_timeseries(ts, variable = "flow_obs", facet = FALSE)

## End(Not run)
```

read_attributes

Read CAMELS-PE Catchment Attributes

Description

Reads catchment attributes from the CAMELS-PE dataset. Attributes are stored by thematic groups, such as topography, climate, hydrological signatures, land cover, geology, soil, and human intervention.

Usage

```
read_attributes(type = "all", gauge_id = NULL, path = get_camels_path())
```

Arguments

type	Character string. Attribute group to read. One of "topographic", "climatic", "hydrological", "landcover", "geologic", "soil", "human_intervention", or "all".
gauge_id	Optional character vector. Gauge IDs used to filter the returned attributes.
path	Character string. Optional path to the CAMELS-PE root directory. If not provided, the path previously defined with <code>set_camels_path()</code> is used.

Details

Available attribute types are:

- "topographic": topographic attributes.
- "climatic": climatic indices.
- "hydrological": hydrological signatures.
- "landcover": land-cover attributes.
- "geologic": geologic attributes.
- "soil": soil attributes.
- "human_intervention": human intervention attributes.
- "all": all available attribute groups joined by `gauge_id`.

Value

A tibble with CAMELS-PE catchment attributes.

Examples

```
## Not run:
set_camels_path("D:/DATA/CAMELS-PE")

# Read topographic attributes
topo <- read_attributes(type = "topographic")

# Read climatic attributes for selected catchments
clim <- read_attributes(
  type = "climatic",
  gauge_id = c("PE_0001", "PE_0002")
)

# Read all attributes
attr_all <- read_attributes(type = "all")

## End(Not run)
```

read_geospatial

Read CAMELS-PE Geospatial Data

Description

Reads geospatial data from the CAMELS-PE dataset, including gauge locations and catchment boundaries. The data are stored as GeoPackage files in the 04_geospatial directory.

Usage

```
read_geospatial(type = c("gauges", "catchments"), path = get_camels_path())
```

Arguments

type	Character string. Either "gauges" or "catchments".
path	Character string. Optional path to the CAMELS-PE root directory. If not provided, the path previously defined with set_camels_path() is used.

Details

Available types:

- "gauges": point locations of gauging stations.
- "catchments": polygon boundaries of catchments.

Value

An sf object.

Examples

```
## Not run:
set_camels_path("D:/DATA/CAMELS-PE")

# Read gauges
gauges <- read_geospatial("gauges")

# Read catchments
catchments <- read_geospatial("catchments")

# Plot
plot(catchments$geometry)
plot(gauges$geometry, add = TRUE, col = "red")

## End(Not run)
```

read_metadata

*Read CAMELS-PE Station Metadata***Description**

Reads the station metadata table from the CAMELS-PE dataset. This file contains the main information for each gauging station, such as the station identifier, name, location, drainage area, river, basin, or other available metadata fields depending on the CAMELS-PE dataset version.

Usage

```
read_metadata(path = get_camels_path())
```

Arguments

path	Character string. Optional path to the CAMELS-PE root directory. If not provided, the path previously defined with <code>set_camels_path()</code> is used.
------	--

Details

The function reads the file:
 01_metadata/stations.csv
 from the CAMELS-PE root directory.

Value

A tibble with CAMELS-PE station metadata.

Examples

```
## Not run:
# Define the CAMELS-PE dataset path
set_camels_path("D:/DATA/CAMELS-PE")

# Read station metadata
```

```
stations <- read_metadata()

# Preview the first rows
head(stations)

## End(Not run)
```

read_timeseries	<i>Read CAMELS-PE time series</i>
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Description

Reads time series data from the CAMELS-PE dataset. The function supports two modes:

Usage

```
read_timeseries(
  gauge_id = NULL,
  global = FALSE,
  vars = NULL,
  path = get_camels_path()
)
```

Arguments

<code>gauge_id</code>	Optional character vector of gauge IDs. Required if <code>global = FALSE</code> .
<code>global</code>	Logical. If TRUE, reads the global file. Default is FALSE (recommended for performance).
<code>vars</code>	Optional character vector of variables to keep.
<code>path</code>	Optional CAMELS-PE root path.

Details

- **Global mode:** reads the full dataset from `timeseries.csv`.
- **By-catchment mode** (recommended): reads only selected catchments from individual files in `by_catchment/`.

Value

A tibble with time series data.

Examples

```
## Not run:
set_camels_path("D:/DATA/CAMELS-PE")

# Read a single catchment
ts <- read_timeseries(gauge_id = "PE_0001")

# Read multiple catchments
ts <- read_timeseries(gauge_id = c("PE_0001", "PE_0002"))
```



```
# Read global dataset
ts_all <- read_timeseries(global = TRUE)

## End(Not run)
```

set_camels_path	<i>Set CAMELS-PE dataset path</i>
-----------------	-----------------------------------

Description

Set CAMELS-PE dataset path

Usage

```
set_camels_path(path)
```

Arguments

path	Character. Path to the CAMELS-PE root directory.
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Value

Invisibly returns the normalized path.

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